

ICS EXE 3

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Problem1: Cache

Consider a **12-bit machine** with a **2-way set** associative cache. The size of each block is **4 bytes**, and the total number of sets is 4. **LRU** policy is used for eviction and **write back** policy is used. The following table shows the content of the data cache at time T.

Set	Tag	Valid	Byte0	Byte1	Byte2	Byte3	Tag	Valid	Byte0	Byte1	Byte2	Byte3
0	0x08	1	0xAC	0xAB	0x08	0x60	0x6A	1	0x11	0x1F	0x64	0x04
1	0xFE	1	0x56	0x01	0x00	0xCC	0x92	0	0xCA	0xBA	0x9F	0xF0
2	0x08	0	0x4C	0xAB	0xFF	0xFF	--	--	--	--	--	--
3	--	--	--	--	--	--	0x92	1	0x93	0x10	0xFD	0xBB

1. What is the total size of the cache? _____ (3')
2. How would a 12-bit physical memory address be split into tag/set-index/block-offset fields in this machine? (6')
tag [1] bits, set-index [2] bits, and block-offset [3] bits
3. A short program will read memory in the following sequences starting from time T. Each access will read **one byte**. Please fill the following blanks. If there is a cache miss, fill '-' for 'Byte Returned'. (10')

Order	Address	Hit/Miss	Byte Returned
1	0x081	[1]	[2]
2	0x92C	[3]	[4]
3	0x680	[5]	[6]
4	0xFE7	[7]	[8]
5	0x6A0	[9]	[10]

Problem 2 Storage Hierarchy

From Figure 6.1 in CSAPP book, we have learned three typical storage devices:

DRAM, SSD and **Hard disk**, in which DRAM is volatile, and SSD and Hard disk are non-volatile (persistent).

1. We know that **random write** is slower than **sequential write** in both **Hard disk** and **SSD**. Are they due to the same reason? If yes, explain the reason. If no, explain the specific two reasons(4').
2. Suppose there is a 16x8 DRAM as figure shows and memory controller sends a request to DRAM to get one byte value.(3'*2=6')

(1) RAS = [1]				
	0	1	2	3
0	A	2	5	7
1	0	5	F	8
2	1	1	A	B
3	B	3	D	A
	-	[3]	-	-
Row buffer				

(2) CAS = [2]				
	0	1	2	3
0	A	2	5	7
1	0	5	F	8
2	1	1	A	B
3	B	3	D	A
	-	-	-	-
Row buffer				

Suppose we finally get value **0xF**, please fill the blank.

3. Recently with appearance of new storage technologies such as PCM(Phase-change memory) and 3D XPoint, a new kind of storage device called **PM(Persistent Memory)** is invented.

It has three main features: **1) Persistence** (持久性): data in it will not be lost after power failure ; **2) Byte-addressability**: it can be accessed like DRAM instead of block devices (SSD, HDD) which can only be r/w at block granularity; **3) Comparable performance** to DRAM: it has less than 10x latency and 1/6-1/3 bandwidth compared to DRAM.

Please choose the most suitable storage device for each application (use **DRAM, PM, SSD** and **Hard disk** to fill the blank). **Note**: each device occurs only once.(4'*2=8')

Fronted cache for backend database	[1]
Video player	[2]
Long-loading games	[3]
Persistent data structures	[4]